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AI based Dynamic Model for Question Paper Generation and Performance Monitoring

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***Abstract*— By generating defined problems out of generic algorithms and assessing the students' abilities at each point of the practice paper, a robust universal platform is created for learning math by personalizing each problem to help the students perform much better by guiding them through. Every problem has a degree of difficulty, which increases when the student solves it successfully. This will allow us to keep track of students' capabilities, and will provide them with practice papers that are based on the difficulty of the previous attempt. The process will start with database generation of Algebra vertical and then building an algorithm to generate the question papers based on students' performance.**

I. INTRODUCTION

Exams play a vital role in a student's life for their capability testing. In the current education system, there is no concept of adaptive learning. Grading systems are often used to score student performance.

One common method is the standardized 10-point scale or to assign a letter grade according to the percentage scored. However, these systems only convey a raw outcome score for the student performance. These systems do not report the level of effort or participation that a given student exhibits.

And hence it leads to grouping the students on basis of different types of grading scales and makes the studies monotonous for students.

It is important to have a smart development question model for growth of students as well as to test learning skills. So, we propose a Question Paper generator that defines the scope of different types of question papers. It defines the degree of difficulty and will test the cognitive skills of the students. Also, Assessment as a dynamic process produces data from which reasonable conclusions are derived. Data mining methodology contributes to proactively ensuring students maximize their academic performance.

The questions recommended to the students will be purely based on his/her previous attempts. Questions will be completely different for every student irrespective of the fact that they would be attempting questions of the same topic. Therefore, it will improve the student performance tracking system (monitoring system) and will also guide them through the process for their skill improvement.

Thus, there will be a personalized attention towards every student and there will be no assumption by the teacher that the students have grasped the particular topic.

II. OBJECTIVE

The main goal of this project is to develop a Robust, open source and free to all platform:

- To develop an open-source Mathematics (Algebra vertical) bilingual app (English and Marathi) for SSC students (grade V to X).
- To improve the student's problem-solving skills by monitoring their performance by providing the handholding.
- To develop interest in students having Mathematics anxiety.
- To develop a system for analyzing the performance at student, class and school level.

III. LITERATURE SURVEY

Information filtering systems that provide a personalized item recommendation to a user in a service environment that can contain or gather varied data are known as recommendation systems. Playlist generators for video and music services, product recommenders for online businesses, content recommenders for social media platforms, and open web content recommenders are all examples of recommender systems in use today.[1] The focus of the educational recommendation system differs from that of the ecommerce, music, and other recommendation systems. Students are given material recommendations, exercises, and tests through the recommendation system in education.[2]

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AI in the education sector has helped us gradually shift towards the digital platform.

The traditional classroom approach is being supplanted by online education. When we think about online education, especially in the context of India, a few platforms come to mind: Udemy, Unacademy, Vedantu, and, of course, the platform that is driving the Indian EdTech market.

TABLE I. TABLE TYPE STYLES

Comparison			
Other Apps	**About**	**Weakness**	**How Eklavya app overcomes these weakness**
Meritnation	This app is completely free and makes all the solved questions from NCERT Books & popular reference books available on your phone.	Question papers are pre-framed and same for everyone.	Questions are recommended on the basis of student's previous attempts
BYJU'S	Byju's has a really wide reach due to all the various courses available on its site. Byju's provides you with courses even for IAS, JEE, NEET and other government exams as well.	It charges around 3,000 per month totals to a whopping amount of 36,000/- per year.	Eklavya app is completely free.
Khan Academy	It is an Interactive platform for students and teachers with over 4000 videos on educational topics like Maths.	It does not focus on individual students and their personal developme nt regarding the subject.	Every student's learning journey is personalized.
Photomath	This is an Artificial Intelligence based application which uses 2 types of Algorithms namely OCR and Equation Solving Programs. User needs to click the Image of the written math problem and the Application returns the solution.	It does not generate questions or question papers for students. Performan ce of students is not taken under account.	Every attempt by the student is monitored and accordingly questions are generated.

IV. PROPOSED METHODOLOGY

Image Stitching generally combines multiple images that have field. The scope of our project is divided into three major phases:

- Database Generation
  1. Generating a database for the Algebra vertical module.
  2. Generated through an algorithm of a generic problem with all sorts of possible variations.
- Designing of an algorithm to generate Question paper

○ Requirement analysis and design a system generated question paper for the following different types of question papers:

■ Practice paper
■ Mock paper
■ Actual paper
■ Competitive paper

○ The algorithm for generating the wrong options should also be structured, based on the knowledge of steps where the students can commit a mistake while solving the problem.

○ For any subject, for any faculty, for any degree of difficulty, these types of questions and types of answers are to be taken under consideration.

○Text, image, audio, video

○ MCQ - singular/multiple correct, Match the pair, True/False, Fill in the blanks,

○ Types of answers and combination of answers along with time given to solve the questions will also decide the degree of difficulty.

Performance Monitoring

○ Identify the attributes to monitor the performance of the student for different types of question papers namely:

■ Practice paper
■ Mock paper
■ Actual paper
■ Competitive paper

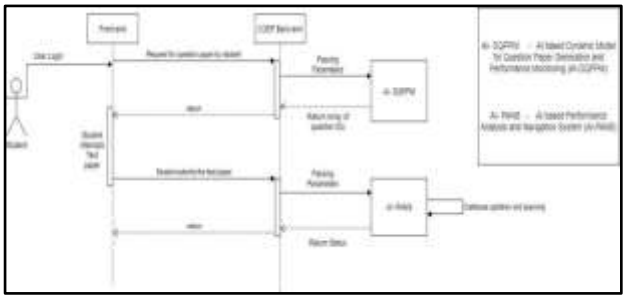

Fig. 1. Proposed Methodology System Diagram

So, when the user logins, he/she will be requesting for the question paper via the front end. The request gets passed on to the back end and after that the parameters are passed to the AI-DQPPM algorithm. Once the AI-DQPPM algorithm functions, it will return an array of questions which the user gets to see at the front end. Hence the user will be attempting the paper and submits the paper. Again, parameters are passed and the AI-PANS

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algorithm updates the database. Hence the final status returns and the user gets to see the updated status on the front end.

V. BLOCK DIAGRAM

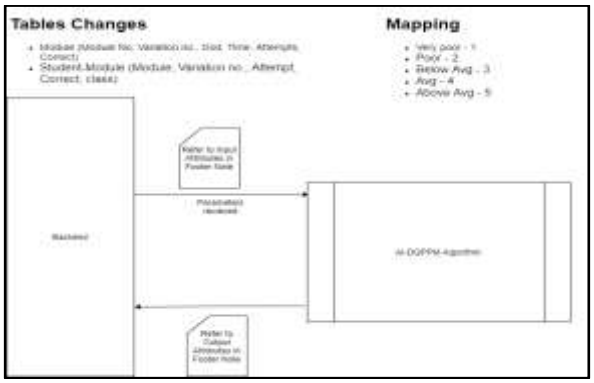

Fig. 2. Implementation
This image gives an overview of the AI-DQPPM algorithm. The method receives parameters from the backend, such as the module number of the array and the StudentID. The output attributes (Question id) are handed on once the algorithm has completed its task.

VI. DATABASE DESIGN AND IMPLEMENTATION

Fig. 3. A snippet of the topics which have been followed for database generation

Data-Bank Structure:
https://docs.google.com/document/d/1Jvz_O6EorA2BpLXhjimhPoTtXlby4us4Qhqa1fZF3Qg/edit?usp=sharing

VII. PROJECT IMPLEMENTATION

A. *CSV Generation*

Master CSV contains attributes as follows - Question Type, Answer Type, Topic Number, Correct Answer(s), Wrong Answer(s), Time taken in seconds, Degree of Difficulty, Question, Question Type, Solution, Solution Type and Variation Number.

All the questions are generated by generic algorithms where no duplicate records are allowed. Latex has also been implemented.

Fig. 4. Snapshot of Excel sheet generated in module 030101

B. *Question Paper Generation System Database:*

- Student Database:
-

student_id	first_name	last_name	email	usern
1	Student	Raina	student@gmail.com	stude
2	hemkesh	raina	hemkeshraina@gmail.com	hemk

Fig. 5. Student Database

- Dynamic Question Database:

qpid	sid	question_id	question	first_answer	second_answer
1	1	571	Following two expressions $11 - 5$ and $1 \times ...	False # चूक	True # बरोबर
1	1	572	Following two expressions 2×4 and $19 - ...	False # चूक	True # बरोबर
1	1	573	Following two expressions $13 - 1$ and $6 \times ...	False # चूक	True # बरोबर

Fig. 6. Dynamic Question Database

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- Student Result Database:

result_id	student_id	question_id	user_answer	right_answer	added_date	qpid
274	1	244	1	2	2021-11-11	1
275	1	251	3	2	2021-11-11	1
276	1	248	1	2	2021-11-11	1
277	1	247	1	2	2021-11-11	1
278	1	249	1	2	2021-11-11	1
279	1	250	3	2	2021-11-11	1
280	1	243	2	2	2021-11-11	1
281	1	246	2	2	2021-11-11	1
282	1	245	1	2	2021-11-11	1
283	1	252	3	2	2021-11-11	1
284	1	254	1	2	2021-11-11	2
285	1	256	1	2	2021-11-11	2

Fig. 7. Student Result Database

- Database Generation:

We have created python scripts for varying types of questions based on algebraic expressions, where we generate multiple questions with random variables and unique constants so as to always generate unique questions as you can see in figure b.1.

```
================================ REST
\Code\0302\030201\03020101\03020101_1
===
The addition of a certain number and
form.<br> # एका संख्येत $9$ मिळवल्यावर उत्तर
हे विधान बरोबर दाखवते?
1 .  $9y=12$
2 .  $y+9=12$
3 .  $9y+9=12$
4 .  $y=9-12$
Answer: $y+9=12$<br>Let's assume that
given that addition of $y$ and $9$ is
ct equation is $y+9=12$<br># उत्तर: $y+
br>या संख्येत $9$ मिळवल्यावर उत्तर $12$ येते अ
करण $y+9=12$<br>
The addition of a certain number and
n form.<br> # एका संख्येत $10$ मिळवल्यावर
ण हे विधान बरोबर दाखवते?
1 .  $c+10=15$
2 .  $10c=15$
3 .  $10c+10=15$
4 .  $c=10-15$
Answer: $c+10=15$<br>Let's assume tha
s given that addition of $c$ and $10$
rrect equation is $c+10=15$<br># उत्तर:
मानू. <br>या संख्येत $10$ मिळवल्यावर उत्तर $15
याचे समीकरण $c+10=15$<br>
```

Figure 8. Output of database generation code

- Student Profile:

Fig. 9. Here the student can check his/her profile details and can also modify it.

- Student Login:

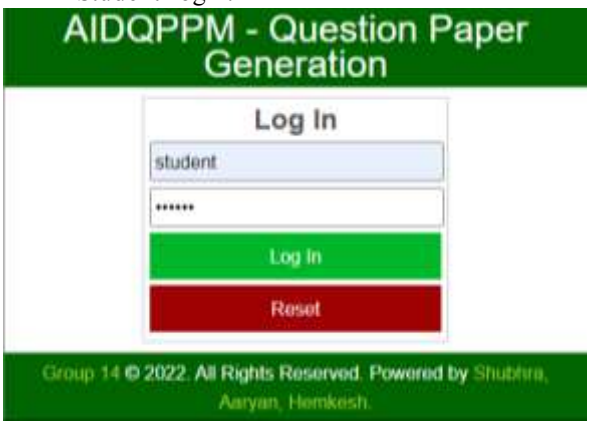

Fig. 10. This is the login interface through which the student can enter the exam portal.

- Curtain Raiser:

The Curtain Raiser is like a module description document containing basic information about the topic of which the student is attempting the test.

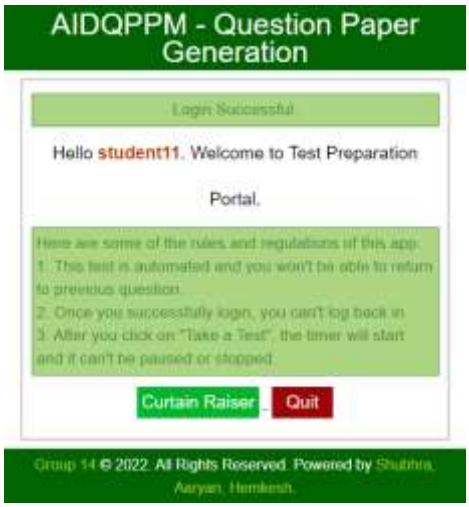

Fig. 11. The student can attempt the test only after he/she has seen the module description as the 'Take a Test' button only appears when the student clicks on the 'Curtain Raiser' button

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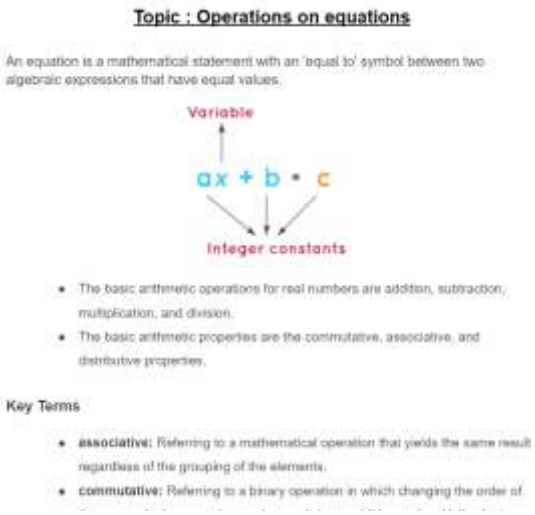

Fig. 12. Since the student has seen the module description. Now he/she can take the test of the selected module.

Fig. 13. Page for student after logging in [6.3]

- Test Portal:

It is a portal that contains the user information at the top, questions in English as well as Marathi at the left side and options of the question on the right side of the Test Portal. It also contains two buttons at the bottom that allows the student to go to the next question and the other to quit the test.

Fig. 14. Test Portal where the student will be attempting the paper

- Pop-up after completion of test:

Here the student can view his/her summarized marks as well as view his entire result and can logout from the system.

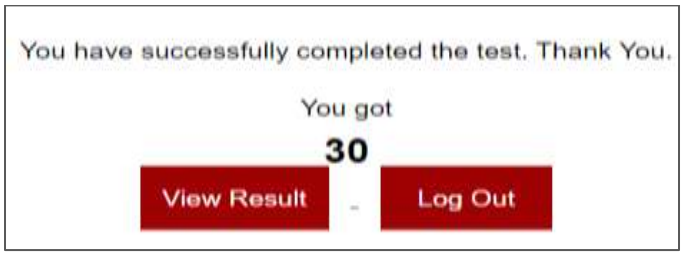

Fig. 15. Pop up for the student once he/she is done with attempting the paper and can view their score as well as result

- Student's Result status:

Here the student can view the entire result containing the questions along with their detailed solutions as well as an indication of whether the student has answered the questions correctly or not (Red - incorrect and Green - Correct)

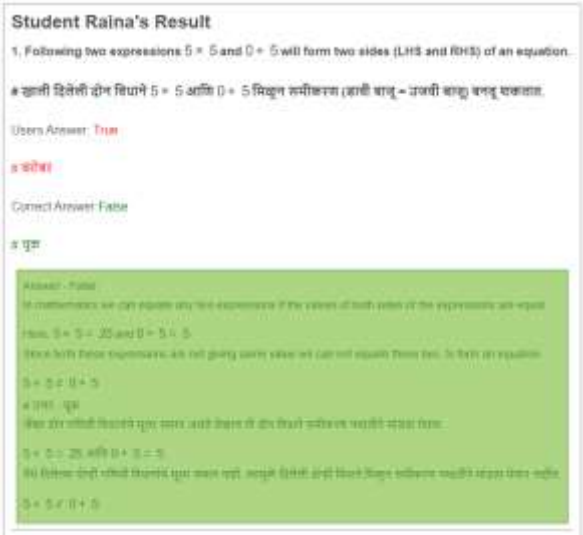

Fig. 16. Student's Result status

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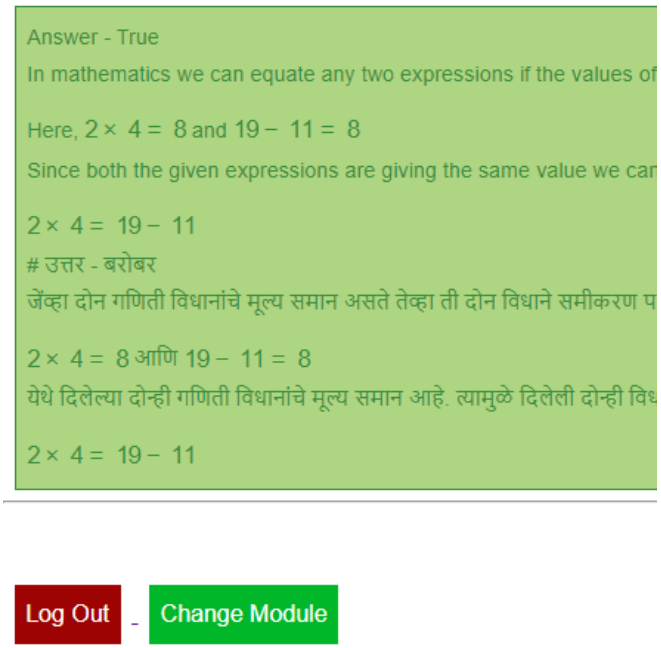

Fig. 17. After viewing the result, the student can either opt to logout or request for manually changing the module for next attempt

Fig. 18. Changing the module manually

VIII. Performance Monitoring & Student Navigation:

- Dynamic Question Paper Recommendation:

The recommendation engine will recommend the number of questions of specific DODs for the student based on the performance of the student in the previous question paper.

```
[[1, 4], [2, 6], [3, 6]]
Recommeding these questions:
4  questions of DOD  1
6  questions of DOD  2
6  questions of DOD  3
```

Fig. 19. This graph shows the correct and wrong percentage of questions answered by a student which comes under the performance monitoring domain.

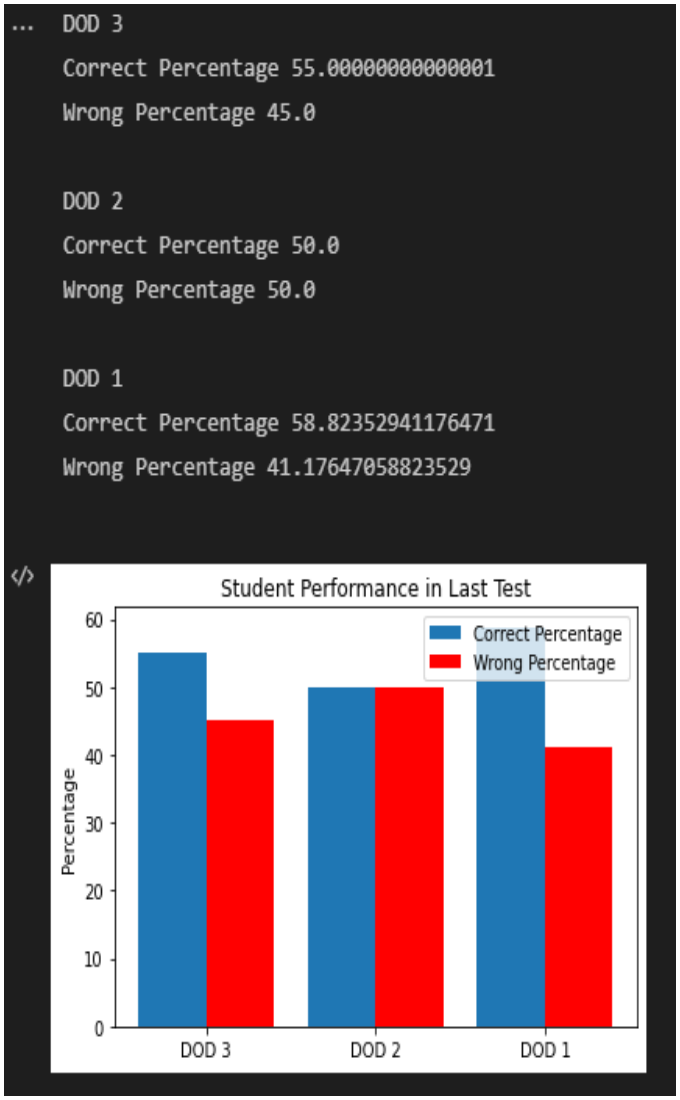

Fig. 20. Graph for student performance represented in the format of graph

- Navigation based on performance analysis:

Based on the performance of the student in the previous question paper, he will be upgraded or downgraded to a different modules. The navigation is based on his pattern of attempting the question of specific DOD clusters.

```
Correct Percentage:
54.54545454545454
Wrong Percentage:
45.45454545454545
Total number of rows in table:  1
downgraded to 30101
MySQL connection is closed
```

Fig. 21. Downgraded

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Recommendation engine will recommend questions based on DODs as well as the previous performance of the student. The number of questions will increase when the student gives poor performance in that specific body student practice more for that DOD the number of questions for dad body will increase when the student give for poor performance and will depend upon how much work performance gives in it for example if he is 10% wrong than the average of all the other duties he will get to more questions if he is 20% wrong he will get for more questions like that we will give him questions based on hairs correct and correct levels

- Result Summary:

The result summary will be stored in the database of every Student in a specific and order-wise manner

summary_id	student_id	marks	added_date
80	1	28.00	2021-11-11

Fig. 22. Result Summary

- DOD Clusters:

The clusters will be based on dynamic Degree of Difficulty (DOD)of the questions which will be based on the number of steps it will take to reach the solution and the average time it will take to attempt the question.

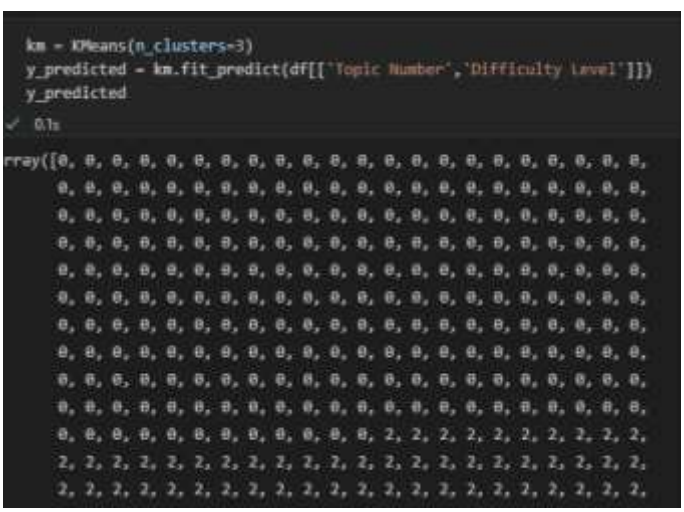

Fig. 23.

- Cluster Centroid:

The graph shown below will represent the centroid of all the DOD based clusters.

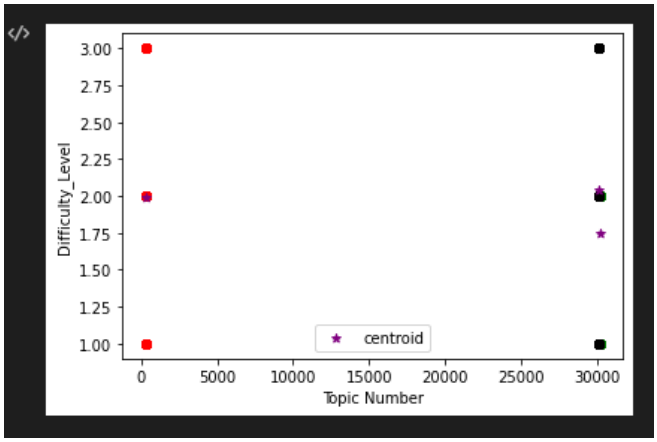

Fig. 24. Graph showing cluster centroids

IX. Additional features

The application will also contain the option of convenience to add the questions and the options manually.

Fig. 25. Manually add questions

X. CONCLUSION

We are working on the Virtual Math-Lab App under the "Virtual Laboratory" project of the Ministry of Education, Govt. of India, in collaboration with COEP. The front end and database will be common between COEP and VESIT colleges. The front end is currently ready at their end, whereas the database will be generated by us for the vertical 3 - Algebra. The backend algorithm will be developed by our team at VESIT using machine learning and AI algorithms and frameworks. This paper presents the design and

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implementation of the database of questions of varying complexity with different degrees of difficulty that will be used in question paper generation.
The system analyses based on prior knowledge i.e., based on his/her performance. Hence, we perform predictive analytics and Automated interventions.

XI. FUTURE SCOPE

The AI-DQPPM is designed keeping in mind many future possibilities that can improve the software to be more productive and secure.

These include:

- We have worked on the Algebra vertical till now, but in the future, we will be working on other verticals as well such as Geometry, Lines and Angles, Trigonometry, Surds and Indices etc.
- We have developed the database questions in text and image format, but we look forward to frame questions in audio as well as video format also.
- The methodologies that we presented for generating question papers show promising outcomes and can be utilized as a basis for making a more advanced automatic and independent paper generator.
- This system can take its true form when a large number of questions are added into a dedicated database which will unlock and also be achieved, which will make sure that only an authorized person can have access to this system.
- Our effort is to develop an open-source Mathematics (Algebra vertical) bilingual app (English and Marathi) for SSC students (grade V to X).
- The key features of our system include Ease of operation as compared to other applications, Analysis based on prior knowledge i.e., based on his/her performance, Automated Navigation and handholding, Predictive analytics, and Automated interventions, Bilingual app (English and Marathi).
- This will be beneficial to the students as they are more inclined to solve multiple papers.
- With all these possibilities, this software will be more secure and will also provide much better results.

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